

# Munawar Kazmi

## Curriculum Vitae

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### Research Interests

Human-robot interaction and trust calibration; legible robot motion; the reliability and verification of language-model planners. I work on how a robot should communicate so that people trust it appropriately, and on how to detect, with machine-checked evidence, when a planner is operating outside what it can do.

### Education

#### MSc Artificial Intelligence and Robotics, Commendation

2023 - 2024

University of Hertfordshire, UK. Thesis: *Enhancing Human-Robot Companionship: Comparing Verbal and Non-Verbal Communication Cues in Humanoid Robots*. Supervisor: Dr Patrick Holthaus.

#### BEng Mechatronics Engineering

2018 - 2022

National University of Sciences and Technology (NUST), Pakistan.

### Publications and Preprints

Kazmi, M. (2026). *Certified Bounds on Achievable Legibility under a Path Cost Budget*. Submitted to IEEE Robotics and Automation Letters, August 2026. Preprint, Zenodo. DOI 10.5281/zenodo.21834955.

Kazmi, M. (2026). *plan-failure-bench: A Machine-Checkable Benchmark of How Language Model Planners Fail*. Working paper, Zenodo. DOI 10.5281/zenodo.21756817.

Kazmi, M. (2025). *Enhancing Human-Robot Companionship: Comparing Verbal and Non-Verbal Communication Cues in Humanoid Robots*. MSc thesis research, University of Hertfordshire.

Kazmi, M. (2026). *The Frozen Blowup Profile of the Okamoto, Sakajo and Wunsch Family: A Control Variate for its Spectral Exponent*. Working note, not yet submitted.

### Research Projects

#### legibility-bounds: certified limits on legible motion

2026 · github.com/munawarkazmi/legibility-bounds

Sole author. Python, computational geometry, exact predicates.

- Proved a ceiling **no trajectory within the path budget can exceed**, where a search can only report the clarity it happened to find. **32 of 32 certified intervals hold with zero violations**; the widest gap is 0.0567 and the narrowest 0.0064.
- The same argument prices safety: **3 of 8 keep-out pairs certify a positive cost**; the other 5 certify nothing and are reported as nothing.
- Stated limit: the observer model is exactly reproducible and has not been validated against people.

#### plan-failure-bench

2026 · github.com/munawarkazmi/plan-failure-bench

Sole author. Python, PDDL, differential testing, CI.

- Designed **60 proof-carrying instructions** over two symbolic environments; every ground-truth label is a mechanical proof re-verified by **589 CI tests** on each commit.
- **Judge-free scoring**: a deterministic checker, differentially tested against an independent PDDL toolchain, assigns one verdict per response. Detection is never reported without its paired false positive count.
- Ran an **18-run grid over four models**, plain and semantically obfuscated; versioning the obfuscation caught and retired two artefacts of its own first token scheme.

#### legible-motion-bench (pilot)

2026 · github.com/munawarkazmi/legible-motion-bench

- Measures legibility against path cost and constraint satisfaction, computed exactly, with no human rater and no model judge. In a three-model pilot the models called **116 of 120** of their own trajectories legible, including all 25 that were not physically possible. One temperature and three models: a pilot, not a finding.

#### Verbal and non-verbal intent communication on the JD humanoid

2023 - 2025

MSc thesis study, University of Hertfordshire. Isaac Sim digital twin, ROS2, physical deployment. Ethics approval cSPECS/CL/UH/05098.

- Built a sim-to-real pipeline so that a gestural and a spoken condition could be delivered identically to **20 participants**, within-subjects and counterbalanced, measured on the Robotic Social Attributes Scale.
- Speech conveyed intent more clearly than gesture (4.9 against 5.8 on a 7-point scale,  $p = 0.017$ ,  $d = 0.58$ ), while engagement and warmth were comparable. Every gestural failure was a failure of joint attention.

## Research Experience

**Robotics Integration Specialist, Robot House, University of Hertfordshire** Jan 2024 - May 2024  
Built the gesture recognition and motion planning pipeline for the JD humanoid in Unity and C++, integrated through ROS2, and deployed PyTorch non-verbal communication models onto its real-time control loop.

**Research Associate, University of Hertfordshire** May 2023 - Dec 2023  
Fine-tuned and optimised TensorFlow models for real-time inference on the Unitree Go1 quadruped, then integrated them into the platform's control loop within the latency budget a walking controller leaves for perception.

## Teaching

**Visiting Lecturer, UCLTILS** Multan, Pakistan; Sep 2026 - present  
Teach Computing Principles and Logic, a first-year module of the University of London BSc Artificial Intelligence: computing systems, propositional and first-order logic, automata, grammars and computability.

**Visiting Lecturer, AI and Robotics, Pak-Turk Maarif Schools** Multan, Pakistan; Aug 2025 - present  
Designed and teach an applied machine learning curriculum for 30+ students with no prior background, taking project teams through training and evaluating their own classifiers.

## Research Software

**llm-nav-shield.** Composes a verifier and a provably correct planner: 38/38 flawed LLM proposals recovered to verified-safe paths, 10/10 correct halts where no safe path exists, zero unsafe trajectories forwarded.

**ROS2 LLM safety verifier.** A deterministic gate between LLM planners and Nav2: 35/35 and 32/32 unsafe proposals caught across two models, with zero misses over 2,071 constructed cases.

**ROS2 path planning.** A\* and D\* Lite as C++20 Nav2 plugins over a ROS-free core; D\* Lite replans 4.3x faster on average over a seeded 200-trial benchmark, with optimality fuzz-validated against Dijkstra over 185,237 replans.

**exact-predicates.** Exact geometric predicates in C++: 657 adversarial cases where floating point is wrong and the exact version is right, each asserted in CI.

**toolcall-contract.** A contract checker for LLM tool calls: of 40 model calls a schema check passes 39 and the contract passes 32.

## Engineering Experience

**Full-Stack Software Engineer (Contract), Safina International** Multan, Pakistan; Jul 2026 - present  
Sole engineer of a live seven-role school management platform: billing, payroll and an append-only audit trail across 33 tables under PostgreSQL row-level security.

**Embedded Systems Engineer (Contract), Muxtronics** Multan, Pakistan; Jan 2025 - May 2025  
Deterministic C++ firmware for real-time image processing on ESP32, and the LoRa mesh protocol that carried alerts off the device, for a commercial intrusion-detection prototype.

**Embedded Systems Intern, National Centre of Robotics and Automation** Mar 2022 - Apr 2022  
Islamabad, Pakistan. Optimised bare-metal C firmware for embedded controllers, validated across 100+ safety-critical test cycles.

## Talks

*HRI and edge AI applications*, AI and Robotics in Education Conference, University of Hertfordshire. Talk with live robot demonstrations of the companionship research.

*Hands-on robotics and machine learning*, Pak-Turk Maarif Schools. Workshop on gesture control, neural networks and real-time deployment, in which teams built and trained working classifiers.

## Service and Memberships

Student Representative, University of Hertfordshire, 2023 - 2024.

Member, University of Hertfordshire Computer Science Chapter (BCS student chapter), 2023 - 2024.

## Technical Skills

**Robotics and systems:** C++20, Python, ROS2 (Nav2, plugin development), differential and property-based testing, CI/CD, Docker

**AI and simulation:** PyTorch, TensorFlow, TensorRT, LLM evaluation and integration, Isaac Sim, MuJoCo, Gazebo, Unity

**Embedded and full-stack:** ESP32, LoRa mesh, bare-metal C/C++, Jetson Nano, React, TypeScript, PostgreSQL

## Referees

Dr Patrick Holthaus, Reader in Interactive Assistive Technology, University of Hertfordshire (MSc supervisor). Further referees and contact details on request.